

Temporal and Pressure Effects: Evidence from NFL Field Goals

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Abstract

This paper examines whether psychological pressure—proxied by dwindling game time in high-stakes situations—affects field goal accuracy in the National Football League (NFL). Using 23,583 field goal attempts from 2000 to 2023, I estimate a logistic regression model that interacts continuous game time remaining with game closeness and quarter. Unlike prior research that relies on binary clutch classifications, this study employs a continuous pressure measure and controls for kick distance, kicker fixed effects, wind speed, and stadium conditions. Results show that in close fourth-quarter situations, every 100-second decrease in remaining game time reduces the probability of a successful kick by 0.33 percentage points. This effect is absent in non-close games, supporting behavioral theories of performance deterioration under pressure. The findings contribute to the literature on stress, decision-making, and sports analytics, and have practical implications for coaching strategy and real-time risk assessment.

KEY WORDS: *Football, Yerkes-Dodson, Behavioral Economics, American Football, Field Goal Kicking*

1 Introduction

In the National Football League (NFL), field goal attempts frequently occur under extreme time pressure, with outcomes often deciding the fate of games—and sometimes entire seasons. Yet, while mechanical determinants of kick success such as distance, angle, and wind are well-studied, the role of psychological pressure remains elusive. Understanding how athletes perform under time constraints not only informs sports strategy but offers insights into decision-making under stress, a central concern of behavioral economics.

Existing research on pressure effects in sports typically defines “clutch” moments using binary indicators (e.g., final two minutes, tie game). However, this coarse classification may overlook how pressure intensifies continuously as time runs out. Prior studies using such binary proxies—such as Clark et al. (2013) and Long (2019)—report mixed or null effects, raising the possibility that methodological limitations, rather than absence of psychological influence, explain these results.

This study contributes by introducing a continuous measure of time pressure and modeling its interaction with game closeness and quarter in a logistic regression framework. Using over 23,000 field goal attempts from 2000–2023, I estimate whether time scarcity in high-stakes moments systematically impairs performance. The model controls for kick distance (in quadratic form), wind, kicker fixed effects, and game conditions. I find that in close fourth-quarter situations, each 100-second decrease in remaining game time is associated with a 0.33 percentage point drop in field goal success probability—a result consistent with pressure-induced performance decline.

These findings contribute to the literature on sports performance, behavioral responses under pressure, and real-time decision-making, with implications extending beyond football to settings such as procrastination, public speaking, and financial

trading.

2 Literature Review

2.1 Psychological and Physiological Theories of Pressure Effects

Early work such as the Yerkes-Dodson Law from Yerkes and Dodson (1908) proposed an inverted U-shaped relationship between arousal and performance, where moderate arousal improves execution but excessive stress impairs it. Building on this, Baumeister and Showers (1988) review the choking under pressure hypothesis, arguing that high-stakes situations interfere with performance through either self-focus—where automatic motor skills become disrupted—or distraction—where attention is divided between task execution and external stimuli.

Yu (2016) extended this line of research with the Stress-Induced Deliberation-to-Intuition (SIDI) Model, proposing that high stress shifts decision-making from deliberative (System 2) to intuitive (System 1) processes. Liu (2024) provided supporting evidence that stress reduces reward evaluation without increasing risk-taking, consistent with automatic, well-rehearsed responses.

These models suggest diverging outcomes under pressure: choking theories emphasize performance decline due to interference, while dual-process models suggest performance may remain stable or improve if tasks rely on automaticity. This theoretical tension motivates the examination of empirical results across sports.

2.2 Empirical Evidence of Pressure Effects in Sports

Price et al. (2010) found that NBA free throw accuracy declined in close game situa-

tions, especially for less-skilled players. Böheim et al. (2018) reported similar declines when players faced larger home audiences. In biathlons, Harb-Wu and Krumer (2019) showed that high-ranked athletes underperformed at home, while lower-ranked competitors did not, suggesting the role of expectation.

Other studies report the opposite. Berger and Pope (2011) showed that basketball teams slightly behind at halftime were more likely to win, consistent with increased motivation under moderate pressure. In contrast, Morgulev and Galily (2018) examined NBA playoff elimination games and found no performance boost, suggesting that extreme pressure does not consistently increase effort or outcomes.

These conflicting results indicate that pressure effects vary by task type, athlete status, and pressure intensity. This variation is relevant to isolated, high-stakes tasks such as NFL field goal attempts.

2.3 Empirical Evidence of Pressure Effects in NFL Field Goals

Several studies have examined whether pressure impacts NFL field goal performance, often reaching null conclusions. Michael (2013) uses logistic regression and random effects models to predict kick success, finding that variables such as crowd noise and game situation do not significantly improve prediction beyond kick distance and environmental factors.

Clark et al. (2013) construct a comprehensive logistic regression model using data from the 2000–2011 NFL seasons. Their analysis incorporates two main approaches to operationalize pressure. First, they include categorical indicators for high-leverage game contexts—such as playoff games, away status, and whether the kicker was “iced” with a timeout—none of which were found to be statistically significant. Second, they test a series of situational pressure measures combining score differential with time

remaining. These variables are grouped into categorical “clutch” scenarios (e.g., “<2 minutes, tie game”) and similarly fail to yield statistically significant effects on field goal success.

Long (2019), applying a Bayesian framework, also reports no consistent effect of pressure once expected kick difficulty is controlled for. Like the earlier studies, Long’s approach relies on discretized proxies for pressure, such as end-of-game situations and postseason indicators.

Collectively, these studies share a common limitation in their reliance on categorical definitions of pressure, which may not fully capture the continuous and dynamic nature of high-stakes moments. Moreover, much of this literature is based on datasets that are now more than a decade old. As both kickers’ performance and analytic tools have advanced, there is a renewed opportunity to reassess the role of pressure using updated data and more granular, continuous measures.

In contrast to previous work, this study introduces a continuous measure of time pressure and its interaction with fourth-quarter, high-leverage situations. The model incorporates kicker fixed effects, environmental controls, and real-time win probabilities to better isolate pressure effects. Results indicate a statistically significant negative relationship between time pressure and kick accuracy in critical moments. These findings suggest that earlier null results may reflect limitations in variable construction rather than the absence of pressure effects and raise the possibility of performance interference even among elite athletes.

3 Data and Empirical Strategy

3.1 Data Analysis

The dataset comprises 23,583 NFL field goal attempts spanning from 2000 to 2023, sourced via the R package NFLFastR and Pro Football Reference. Each observation includes game time remaining, kick distance, environmental conditions, and outcomes, allowing for detailed analysis of how situational pressure influences kicker performance. This structure enables the construction of interaction-based empirical models to isolate the impact of time pressure.

Table 1 presents descriptive statistics for key variables.

Table 1: Summary Statistics

Variable	Mean	SD	Min	Max	N
Field goal made	0.85	0.36	0	1	23,583
Kick distance (yards)	37.20	10.20	18	76	23,583
Game seconds remaining	1687.00	979.00	0	3600	23,583
Absolute score differential	0.53	9.30	0	45	23,583
Close game	0.71	0.46	0	1	23,583
Wind speed (mph)	6.03	5.89	0	71	23,583

Notes: Sample includes 23,583 NFL field goal attempts from 2000–2023. “Field goal made” is coded 1 if the kick was successful. “Close game” equals 1 if the absolute score differential is less than or equal to 9 points. Kick distance is in yards. Wind speed is recorded at the time of the attempt.

The average field goal success rate is 84.6%, consistent with historical norms in the NFL. The average kick distance is 37.2 yards, and mean wind speed is 6.03 mph. Close-game scenarios (see Table 1 footnote for definition) account for approximately 71% of all attempts.

These figures highlight that field goals are generally successful and often occur in competitive contexts, supporting the focus on how pressure interacts with marginal game conditions. The standard deviation in game seconds remaining is substantial,

allowing sufficient variation to identify pressure effects over time. Wind conditions also vary meaningfully, motivating its inclusion as a control in later specifications.

Table 2 shows the distribution of game quarters, stadium types, and surface conditions.

Table 2: Distribution of Categorical Variables

Game Quarter	Count	Percent
1st Quarter	4459	18.9
2nd Quarter	8507	36.1
3rd Quarter	4457	18.9
4th Quarter	6160	26.1
Stadium Roof Type		
Closed	2047	8.7
Dome	3765	16.0
Open Roof	470	2.0
Outdoor	17,301	73.4
Playing Surface Type		
Grass	13,319	56.5
FieldTurf	6081	25.8
AstroTurf	1291	5.5
SportTurf	1129	4.8
MatrixTurf	635	2.7
AstroPlay	448	1.9
A_Turf	321	1.4
DessoGrass	194	0.8
Unknown / Blank	165	0.7

Notes: Surface and roof types are based on metadata from NFLFastR. Percentages reflect the share of total field goal attempts by category. Percent values are rounded to one decimal place.

Most kicks occur during the second and fourth quarters, with the latter being the primary focus of pressure dynamics. Approximately 73% of attempts take place in outdoor stadiums, and 56.5% occur on natural grass.

Variation in stadium roof and playing surface types also supports the use of envi-

ronmental fixed effects. Field conditions can influence both actual kick difficulty and kicker confidence, and thus must be accounted for when isolating pressure effects.

Table 3 reports kicker-level variation in the number of field goal attempts.

Table 3: Kicker Summary Statistics

Statistic	Value
Number of unique kickers	179
Mean field goals attempted	132.2
Standard deviation	150.0
Minimum attempts	1
Maximum attempts	617

Notes: Distribution of field goal attempts across individual kickers. Used to justify inclusion of kicker fixed effects in regression analysis.

Across 179 unique kickers, the average number of attempts is 132.2, though the range spans from 1 to 617.

This heterogeneity underscores the importance of controlling for individual skill using kicker fixed effects. Without doing so, observed variation in success rates under pressure could conflate psychological effects with differences in baseline ability across kickers.

To assess whether field goal attempts in close games differ systematically in underlying team strength, I conducted a balance test on average team quality using Expected Points Added (EPA) per team-season. Figure 1 and Figure 2 display the distributions of average EPA for close ($|\Delta\text{Score}| \leq 9$) and non-close games. Both distributions appear approximately normal and centered around zero. A two-sample t -test comparing the means yields a p -value of 0.075 with a 95% confidence interval of $[-0.251, 0.012]$, failing to reject the null hypothesis of equal means.

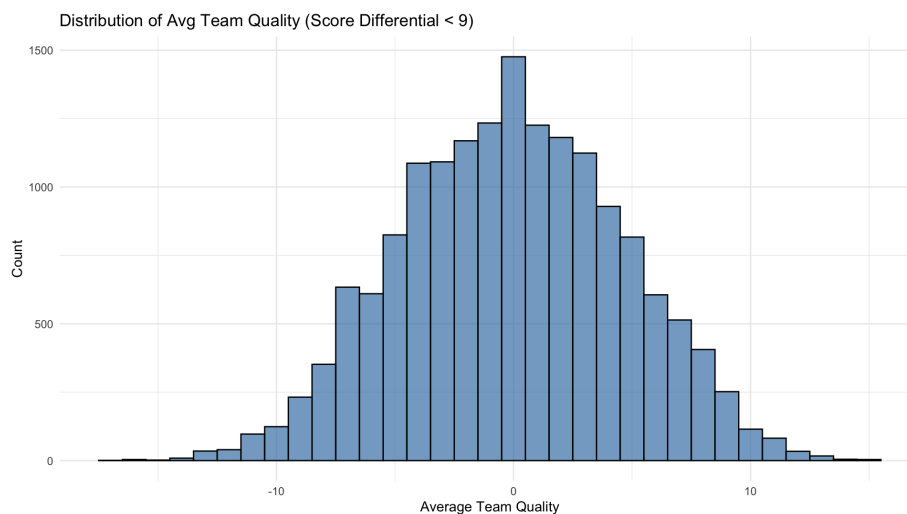


Figure 1: Distribution of Team EPA for Close Games ($|\Delta\text{Score}| \leq 9$)
Notes: Team quality measured using average Expected Points Added (EPA) from season performance.

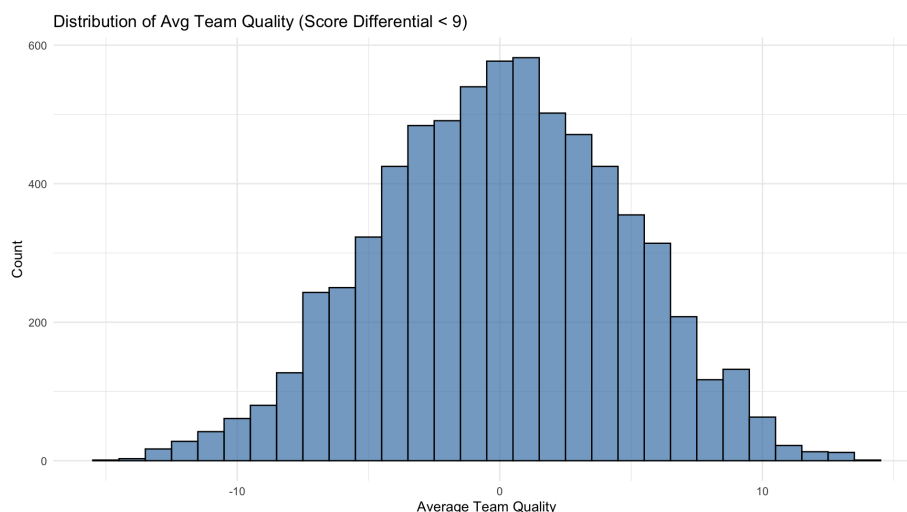


Figure 2: Distribution of Team EPA for Close Games ($|\Delta\text{Score}| > 9$)
Notes: Team quality measured using average Expected Points Added (EPA) from season performance.

These results suggest that game closeness is not strongly confounded by differences in team strength, reducing concerns about selection bias and strategic or endogenous assignment of field goal attempts. The 9-point cutoff is therefore valid for defining comparable competitive pressure environments.

Field Goal Success by Game Time Elapsed

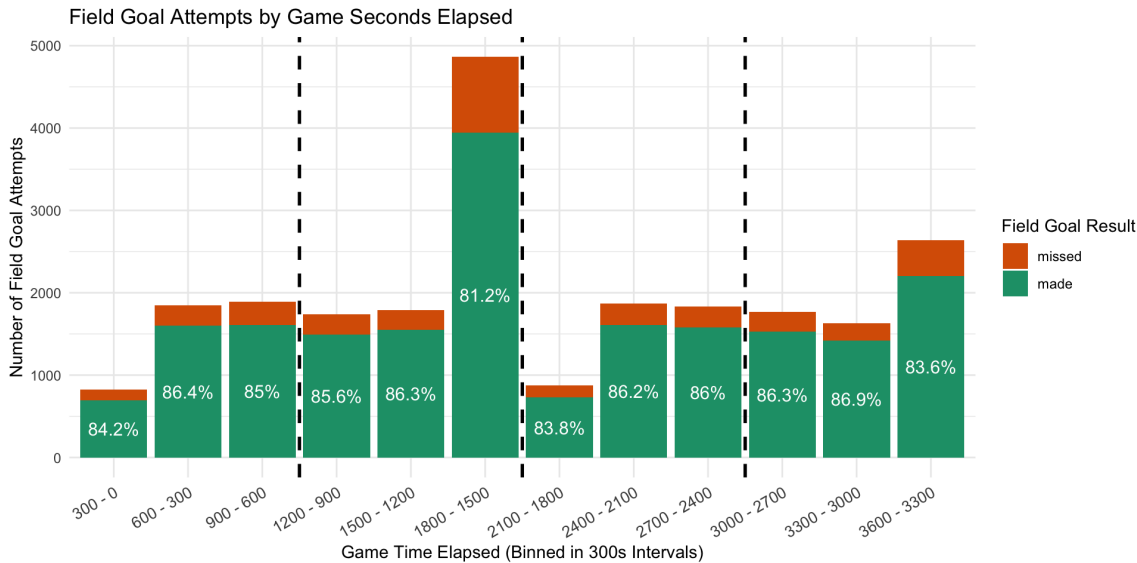


Figure 3: Field Goal Success Rate by Game Time Elapsed

Figure 3 plots field goal success rates across 300-second intervals of game time, distinguishing made and missed attempts. Success rates are relatively stable across most time windows, hovering between 83% and 86%, suggesting that time alone does not drastically impair performance. However, two modest declines are observed. The first occurs between 2400 and 1800 seconds (mid-second quarter), where accuracy drops to 81.3%, possibly reflecting strategic long-distance attempts before halftime. The second occurs in the final 300 seconds of play, where success declines slightly to 83.6%, a dip that may indicate subtle effects of time pressure during high-stakes moments.

These patterns motivate the need to model time as a continuous variable and to investigate whether declines under time constraints interact specifically with game closeness and quarter—rather than appearing uniformly throughout the game.

Field Goal Success by Kick Distance

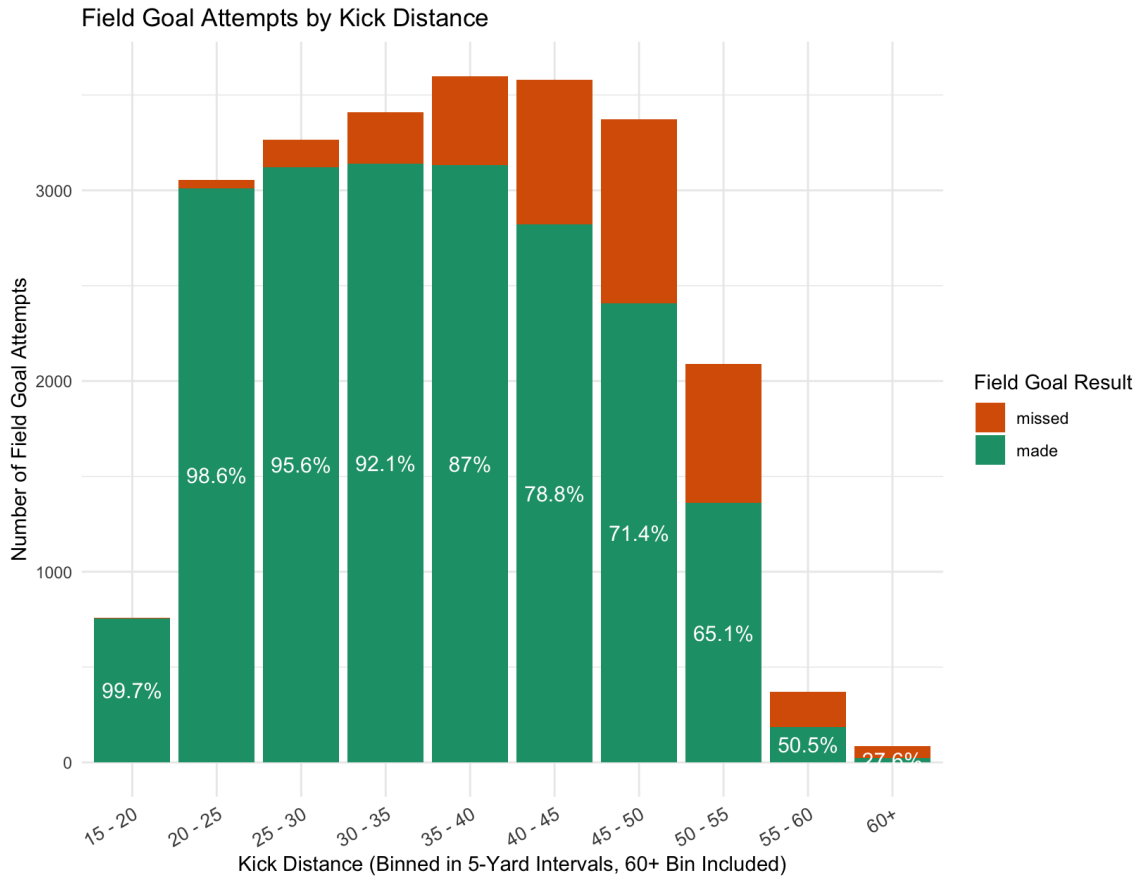


Figure 4: Field Goal Success Rate by Kick Distance

Figure 4 presents field goal success rates by kick distance, grouped in five-yard intervals. Accuracy exceeds 95% for kicks under 30 yards and declines progressively as distance increases. A notable threshold appears beyond 40 yards: success falls to 78.8% at 40–45 yards and further to 71.4% at 45–50 yards. The decline becomes especially sharp for attempts beyond 55 yards, which show only a 50.5% success rate, dropping to 27.6% for attempts over 60 yards.

These results confirm that distance is the strongest mechanical predictor of success and justify its inclusion in both linear and quadratic form in the empirical model.

They also underscore the importance of controlling for environmental and situational factors, which may interact differently across kick distances.

3.2 Empirical Strategy

This study evaluates the effect of time pressure on field goal success by modeling performance as a function of remaining game time, with particular attention to attempts occurring in close games during the fourth quarter. The main explanatory variable is a three-way interaction term combining game closeness, fourth-quarter timing, and continuous time remaining.

Prior studies often define pressure using binary or categorical variables. This paper departs from that approach by constructing a continuous measure of time remaining, which allows for variation in pressure intensity within close-game scenarios. This framework provides a more detailed estimate of how pressure evolves as the game progresses.

The interaction term is constructed using three variables. The first is an indicator for whether the absolute score differential is nine points or fewer (*Close_Game*). The second is an indicator for whether the attempt occurred in the fourth quarter (*Post4*). The third is the number of seconds remaining in the game at the time of the attempt (*Game_Seconds_Remaining*). The dependent variable is a binary indicator equal to one if the field goal attempt was successful.

The model includes covariates to account for observed sources of variation in difficulty and context. These include kick distance, entered in both linear and quadratic form, wind speed, kicker fixed effects, and stadium characteristics such as playing surface and roof type.

The identification assumption is that, conditional on these controls, the precise

timing of a field goal attempt within close games is exogenous to unobserved determinants of success. A balance test using team-level expected points added (EPA) confirms that team strength is similar across close and non-close games, supporting the validity of the game closeness definition.

A logistic regression model is used given the binary outcome. While a probit model provides similar fit, the logit specification is preferred for its interpretability and its suitability in settings where outcome probabilities are skewed toward one, as is the case with NFL field goals.

To assess potential multicollinearity among regressors, I compute Generalized Variance Inflation Factors (GVIFs), adjusting for multi-parameter variables using the transformation $GVIF^{1/(2 \times Df)}$ as recommended by Fox and Monette (1992). As shown in Appendix Table 5, all adjusted GVIF values remain well below the commonly cited threshold of 10, and the main interaction term of interest exhibits a $GVIF^{1/(2 \times Df)}$ below 3, suggesting that multicollinearity is not a substantial concern in the estimation.

The empirical analysis tests the hypothesis that time pressure impairs field goal performance in high-stakes moments. Specifically, the model examines whether remaining game time negatively affects success rates, but only in close-game, fourth-quarter scenarios. The null hypothesis is that remaining time has no differential effect on performance, conditional on game context. The alternative hypothesis is that success probabilities decline as time runs out, but only when the score is close and the game is nearing its end.

4 Model

4.1 Assumptions

To identify the causal effect of psychological pressure on field goal performance, several assumptions are required. First, the timing of a field goal attempt within the fourth quarter of a close game must be exogenous to unobserved determinants of success, conditional on a rich set of controls. Specifically, after accounting for kicker fixed effects, kick distance, environmental conditions, and game context, the remaining game time is assumed to proxy for increasing psychological pressure rather than capturing unobserved factors such as team morale, hidden injuries, or strategic manipulation. This assumption is consistent with performance theories such as the Yerkes-Dodson Law and choking under pressure, which suggest that pressure builds naturally as time runs out.

Second, all variables used in the analysis—including field goal success, game seconds remaining, score differential, and distance—are assumed to be measured accurately. Any measurement error is assumed to be classical and uncorrelated with regressors. Third, each field goal attempt is treated as an independent observation, such that the outcome of one attempt does not affect another either within the same game or across games. This ensures that performance pressure is internally experienced and not a reflection of team-level contagion effects.

While strategic behavior may influence game dynamics, I assume that the timing of field goal attempts is not systematically driven by unobservable kicker-specific psychological states or internal coaching evaluations. Observable features such as game situation and distance largely determine when a kick is attempted, and kicker fixed effects help absorb any remaining variation tied to skill or form. Furthermore, coaches' decisions about whether and when to attempt a field goal are assumed to be

orthogonal to unobserved factors that also influence success. Although some strategic manipulation (e.g., centering the ball or calling a timeout) may occur, these adjustments are presumed infrequent or random with respect to psychological performance effects.

Finally, the model assumes that the logit specification accurately captures the interaction between remaining time, closeness of the game, and quarter effects. The three-way interaction term is intended to represent the nonlinear nature of pressure effects. Importantly, I assume that psychological pressure effects are not systematically compensated for—via timeouts, prolonged preparation routines, or opponent penalties—so that the observed variation in remaining time reflects meaningful differences in the level of pressure faced by the kicker.

4.2 Model Specification

This specification builds directly on the identification strategy outlined in Section 3.2, translating the hypothesized pressure effects into a formal empirical model. Let $Y_i \in \{0, 1\}$ denote whether field goal attempt i was successful. I estimate a logistic regression model where the probability of success is given by:

$$\Pr(Y_i = 1 \mid \cdot) = \Lambda(\alpha + \beta_1 C_i + \beta_2 Q_i + \beta_3 T_i + Z_i \delta + \beta_7 C_i Q_i T_i + X_i \gamma) \quad (1)$$

Here, $\Lambda(z) = \frac{1}{1+e^{-z}}$ denotes the logistic cumulative distribution function. The variable C_i is an indicator for a close game (e.g., score differential within 9 points), Q_i indicates whether the kick occurred in the fourth quarter, and T_i captures the number of seconds remaining in the game. The term $C_i Q_i T_i$ is the three-way interaction that captures the hypothesized pressure effect under high-stakes moments.

To reduce dimensional clutter, I define $Z_i = [C_i Q_i, C_i T_i, Q_i T_i]$ as the vector of all

two-way interactions among C_i , Q_i , and T_i , with corresponding coefficient vector δ . The covariate vector X_i includes wind speed, kick distance (and its square), as well as fixed effects for the kicker and stadium.

The primary coefficient of interest is β_7 , which identifies whether the marginal effect of time remaining on kick success differs in situations that are both close and in the fourth quarter—conditions associated with elevated psychological pressure. Including all two-way interactions in Z_i allows the model to flexibly account for heterogeneous effects without conflating them with the unique three-way term.

5 Results

5.1 Main Effects of Time Pressure

The Table 4 shown below reports the results from the preferred specification, which includes kicker fixed effects, stadium surface and roof type controls, and wind speed as a continuous covariate.

The model estimates the marginal effect of time pressure in close fourth-quarter situations via a three-way interaction term: *Close Game* $\times$ *4th Quarter* $\times$ *Seconds Remaining*. Coefficients are rescaled to reflect marginal effects per 100 seconds remaining.

The estimated marginal effect is 0.003 ($p < 0.05$), indicating that for every additional 100 seconds on the game clock, the probability of a successful field goal increases by approximately 0.33 percentage points in close fourth-quarter scenarios. This result is consistent with the hypothesis that kicker performance deteriorates modestly under escalating time pressure in high-stakes contexts.

Importantly, the other two-way interaction terms—*Close Game* $\times$ *Seconds Remaining* and *4th Quarter* $\times$ *Seconds Remaining*—are statistically insignificant. This supports the interpretation that the observed pressure effect is driven by the conjunction of high-stakes and time scarcity, rather than either factor alone.

Table 4: Regression Results for Field Goal Success (Logit Model, Including Wind)

Variable	Dependent Variable: Field Goal Success			
	(1) Base Model	(2) Kicker FE	(3) Stadium FE	(4) Full Model
Main Effects				
Intercept	6.74*** (0.21)	6.79*** (0.25)	6.64*** (0.25)	6.67*** (0.30)
Close Game (C)	-0.20 (0.19)	-0.15 (0.19)	-0.18 (0.19)	-0.13 (0.20)
4th Quarter (Q)	0.12 (0.21)	0.11 (0.22)	0.11 (0.22)	0.11 (0.22)
Seconds Remaining (T)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Kick Distance	-0.12*** (0.00)	-0.12*** (0.00)	-0.12*** (0.00)	-0.13*** (0.00)
Kick Distance Squared	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)
Wind	-0.03*** (0.00)	-0.03*** (0.00)	-0.02*** (0.00)	-0.02*** (0.00)
Interaction Terms				
C × Q	-0.10 (0.24)	-0.11 (0.25)	-0.10 (0.24)	-0.12 (0.25)
C × T	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Q × T	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
C × Q × T (per 100s)	0.07* (0.03)	0.06* (0.03)	0.06* (0.03)	0.06* (0.03)
Model Fit				
Observations	23,582	23,582	23,582	23,582
Residual Deviance	17,275	16,792	17,125	16,706
AIC	17,297	17,170	17,263	17,200
Fisher Scoring Iter.	6	13	6	13

Notes: Standard errors in parentheses. All models use a binomial logistic regression (logit link). Column (1) adds wind as a continuous predictor. Column (2) includes wind and kicker fixed effects. Column (3) includes wind and stadium fixed effects. Column (4) includes all covariates and fixed effects. **The coefficient for the three-way interaction term (C × Q × T) has been scaled by 100 to reflect change per 100 seconds. This transformation follows: $\hat{\beta}_{100} = \hat{\beta}_1 \times 100$, and similarly for its standard error.**

Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Abbreviations: C = Close Game, Q = 4th Quarter, T = Seconds Remaining.

Figure 5 illustrates the marginal effect of time elapsed in the fourth quarter on predicted field goal success, conditional on game status.

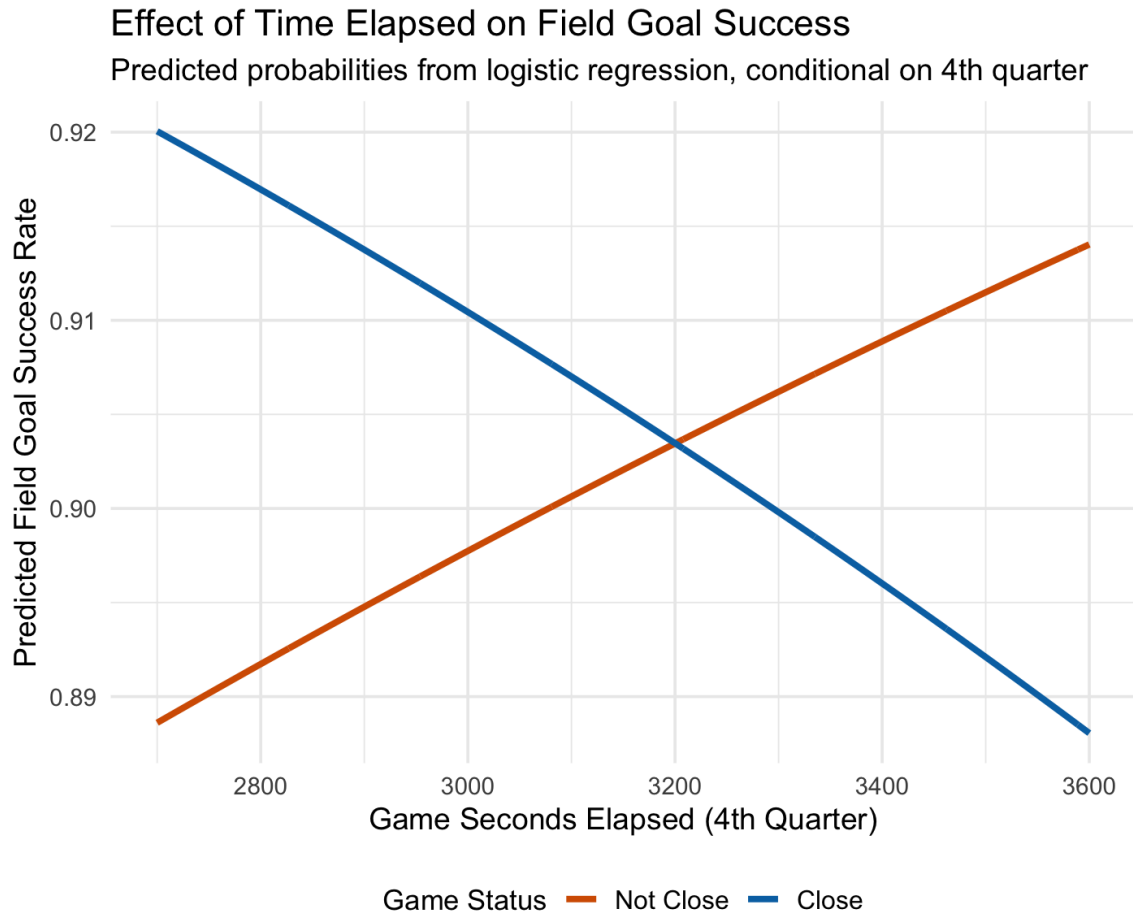


Figure 5: Predicted Success Rate by Kick Distance

In close games, the predicted success rate declines steadily as time elapses—consistent with the hypothesis that pressure builds and impairs performance. Conversely, in non-close games, success rates slightly increase with time, suggesting that in the absence of psychological pressure, kickers may benefit from more favorable field positioning or reduced defensive urgency. The interaction effect is visually evident: as the game approaches its final moments, kickers in close scenarios exhibit a pronounced drop in predicted success compared to their non-close counterparts; an almost 3% drop in field goal performance from the start of the quarter to the final seconds of the game.

5.2 Robustness and Specification Checks

The results are robust across a range of alternative specifications, including models that exclude wind, omit fixed effects, or isolate only the three-way interaction term. Across these variations, the magnitude and direction of the time pressure effect remain stable, with marginal effects ranging from 0.002 to 0.003 per 100 seconds. Appendix Tables 6–9 provide full regression results for these specifications.

The inclusion of kicker fixed effects addresses concerns about unobserved skill heterogeneity, while stadium and environmental controls mitigate confounding from venue-specific conditions. Wind speed is negatively associated with kick success and statistically significant at the 1 percent level, but does not meaningfully affect the estimated impact of time pressure.

Additionally, to assess the stability and credibility of the main findings, I conduct a series of robustness checks using alternative model specifications and sub-sample analyses.

Appendix Table 7 introduces a restricted two-minute-window specification, estimating the same core logistic regression but focused solely on high-pressure attempts occurring within the final 120 seconds of regulation. This restricted window intensifies the contextual pressure and serves as a stress test for the generalizability of time pressure effects.

Across all three columns of Table 7—Base (Column 1), Rookie Kicker (Column 2), and Playoff-adjusted model (Column 3)—the negative effect of time pressure in close-game scenarios remains consistent. The key interaction term, *Close Game* $\times$ *2_Min Indicator* $\times$ *Time Remaining*, is statistically significant in all models (e.g., 1.38*, 1.31, and 1.57*, respectively), reinforcing the interpretation that kickers underperform in the final moments of close games. This result is consistent with the broader three-way

interaction estimates found in the full sample (see Table 4).

Importantly, the effect of time pressure remains directionally stable even after accounting for potentially confounding variables. The inclusion of a *Rookie Kicker* dummy does not meaningfully alter the time-pressure coefficient, suggesting that inexperience is not driving the main result. Similarly, Column 3 incorporates a *Playoff* indicator and four-way interaction, finding that playoff-specific effects do not significantly alter the magnitude or direction of the main pressure effect—though the wide confidence interval on the four-way term reflects limited statistical power in this subsample.

The coefficient on *Seconds Remaining* remains near zero across all models, reaffirming that time itself does not affect field goal outcomes unless compounded with situational pressure (i.e., closeness and game stage). Control variables such as *Kick Distance*, *Kick Distance Squared*, and *Wind* retain strong statistical significance and the expected signs, providing reassurance that the model accurately accounts for mechanical and environmental difficulty.

Taken together, these robustness checks bolster confidence in the main result: kicker performance modestly deteriorates under heightened psychological pressure in close, late-game moments. The result survives alternate model definitions, subsample restrictions, and interaction decompositions, affirming its validity and reinforcing its implications for both behavioral theory and in-game decision-making.

5.3 Economic and Strategic Implications

While modest in magnitude, the estimated decline in field goal success under time pressure has meaningful implications in competitive games, where outcomes often hinge on a single possession. These results align with behavioral theories, such as

the Yerkes-Dodson Law, which predict reduced performance at high arousal levels. Coaches and analysts may need to consider not only field position and kicker ability, but also how dwindling time may subtly erode performance under pressure.

6 Conclusion

This paper investigates whether psychological pressure—proxied by dwindling game time in high-stakes situations—affects field goal performance in the National Football League (NFL). Using a logistic regression framework and over 23,000 field goal attempts from 2000 to 2023, I find that the probability of a successful field goal declines modestly but significantly as time runs out in close, fourth-quarter games. Specifically, each 100-second decrease in remaining game time is associated with a 0.33 percentage point drop in success probability under pressure, after controlling for kick distance, environmental conditions, and kicker ability.

These findings provide empirical support for performance theories such as the Yerkes-Dodson Law and the Choking Under Pressure hypothesis, which predict that excessive arousal impairs task execution. The results also speak to the nature of skill execution under stress: even for highly trained NFL kickers, performance appears susceptible to time-sensitive psychological constraints. In contrast to prior studies that relied on binary “clutch” indicators and found null results, the use of a continuous pressure metric allows for finer detection of stress-performance dynamics. This suggests that earlier null findings may have stemmed from limitations in model specification rather than an absence of pressure effects.

Despite these contributions, the study has limitations. Most notably, pressure is inferred from game context rather than measured directly. Although kicker fixed effects and environmental controls reduce confounding, unobserved factors, such as

coaching strategy, fatigue, or the kicker’s psychological profile, may still bias results. The assumption of exogenous kick timing within close games, while supported by balance tests, remains an identifying assumption rather than a tested fact. Moreover, performance may be shaped by compensating mechanisms like timeouts or prolonged preparation, which this model does not explicitly capture.

These findings raise several avenues for future research. Richer data, such as biometric feedback or in-game audio/video cues—could help directly measure psychological arousal and supplement structural models of performance. Investigating pressure effects across other domains of elite performance, such as soccer penalty shootouts, basketball free throws, or even financial trading environments, may further refine our understanding of how time constraints distort judgment and execution.

In sum, this study offers robust evidence that time pressure affects field goal accuracy in high-stakes NFL scenarios. By moving beyond binary clutch classifications and adopting a continuous framework, it highlights the nuanced ways in which pressure operates, even among expert performers. These insights not only deepen our understanding of athletic performance under stress but also contribute more broadly to behavioral economics and the modeling of decision-making under pressure.

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A Assumptions

Assumption A1 (Exogeneity of Game Timing). Conditional on included controls—such as kicker fixed effects, kick distance, environmental variables, and game context—the exact timing of a field goal attempt within the fourth quarter of a close game is assumed to be exogenous to unobserved determinants of success other than pressure-related psychological effects. That is, remaining game time is not systematically confounded by unobserved factors such as team morale shifts, coach selection strategies, or hidden field conditions. Rather, it functions as a proxy for increasing psychological pressure, consistent with performance theories such as the Yerkes-Dodson Law and choking under pressure.

Assumption A2 (Measurement Validity). All key variables (i.e field goal outcome, game seconds remaining, kick distance, and score differential) are measured accurately and consistently across games and seasons. Any remaining measurement error is assumed to be classical and uncorrelated with the regressors.

Assumption A3 (Stable Unit Treatment Value - SUTVA). Each field goal attempt is an independent unit of observation; no kicker’s outcome is affected by other simultaneous or past attempts within the same game or across games. This assumption ensures that pressure is internally experienced and does not reflect broader team dynamics or contagion effects.

Assumption A4 (No Strategic Sorting). Conditional on game context (kick distance, time remaining, score differential), the timing of field goal attempts within close fourth-quarter situations is not systematically driven by unobserved kicker-specific factors such as psychological state or internal coaching evaluations. While strategic delays may occur, they are overwhelmingly determined by observable game features which are explicitly controlled for in the model. Kicker fixed effects further

absorb selection based on perceived ability or form.

Assumption A5 (Model Correctness and Functional Form). The logit specification correctly captures the functional relationship between the interaction of time, closeness, and quarter effects and field goal success. In particular, the three-way interaction is assumed to appropriately model the nonlinear nature of pressure effects.

Assumption A6 (Invariance to Coaching Behavior). Coaches' decisions regarding the timing of field goal attempts are not systematically correlated with unobserved kicker-specific traits that also affect performance. While coaches may occasionally adjust for game dynamics, these adjustments are assumed to be orthogonal to unmeasured kicker performance determinants.

Assumption A7 (Time Pressure Effects Are Identifiable). The psychological effect of time pressure is not fully offset by compensating behaviors such as timeouts, prolonged routines, or defensive penalties. Such mechanisms are assumed to be infrequent or uncorrelated with the treatment variable (remaining time), allowing the model to detect net performance effects due to pressure.

B Tables

Table 5: Variance Inflation Factors (VIF) for Logit Model Predictors

Variable	GVIF	Df	GVIF ^{1/(2×Df)}
close_game	20.424	1	4.519
post4	24.022	1	4.901
game_seconds_remaining	22.100	1	4.701
wind	1.146	1	1.070
kick_distance_centered	1.658	1	1.288
kick_distance_squared	1.657	1	1.287
factor(kicker_player_id)	831.167	173	1.020
factor(stadium_id)	771.654	39	1.089
close_game:post4	22.082	1	4.699
close_game:game_seconds_remaining	38.856	1	6.233
post4:game_seconds_remaining	9.183	1	3.030
close_game:post4:game_seconds_remaining	7.631	1	2.762

Table 6: Effect of Time Pressure in Close Game Situations (Three-Way Interaction Only)

Variable	Dependent Variable: Field Goal Success			
	(1) Base Model	(2) Wind + Kicker FE	(3) Wind + Stadium FE	(4) Full Model
Main Effects				
Intercept	6.64*** (0.13)	6.73*** (0.19)	6.54*** (0.18)	6.62*** (0.24)
Wind	-0.03*** (0.00)	-0.03*** (0.00)	-0.02*** (0.00)	-0.02*** (0.00)
Kick Distance	-0.12*** (0.00)	-0.12*** (0.00)	-0.12*** (0.00)	-0.13*** (0.00)
Kick Distance Squared	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)
Three-Way Interaction (per 100s)				
CG × Q4 × SR	0.03** (0.01)	0.03* (0.01)	0.03* (0.01)	0.03* (0.01)
Model Fit				
Observations	23,582	23,582	23,582	23,582
Residual Deviance	17,282	16,797	17,131	16,711
AIC	17,292	17,163	17,257	17,193
Fisher Scoring Iter.	6	13	6	13

Notes: Standard errors in parentheses. All models use a binomial logistic regression (logit link). The three-way interaction term (CG × Q4 × SR) is scaled per 100 seconds for interpretability: Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 7: Regression Results for Field Goal Success (Logit Model, 2-Minute Window)

Variable	Dependent Variable: Field Goal Success		
	(1) 2_min Base	(2) Rookie Kicker	(3) Playoff
Main Effects			
Intercept	6.64*** (0.25)	6.68*** (0.25)	6.68*** (0.25)
Close Game (C)	0.08 (0.10)	0.09 (0.11)	0.05 (0.10)
2_Min Indicator	1.00* (0.48)	0.91 (0.52)	1.15* (0.51)
Rookie Kicker		-0.26 (0.23)	
Playoff			-0.95* (0.47)
Seconds Remaining (T)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Kick Distance	-0.13*** (0.00)	-0.13*** (0.00)	-0.13*** (0.00)
Kick Distance Squared	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)
Wind	-0.02*** (0.00)	-0.02*** (0.00)	-0.02*** (0.00)
Interaction Terms			
C × 2_Min	-1.37** (0.50)	-1.29* (0.54)	-1.51*** (0.52)
2_Min × T (per 100s)	-1.11† (0.66)	-1.07 (0.72)	-1.33** (0.68)
C × Playoff			1.02† (0.59)
C × T × 2_Min	1.38* (0.70)	1.31 (0.76)	1.57** (0.71)
C × T × 2_Min × Playoff			-1.45 (18.68)
Model Fit			
Observations	23,582	23,582	23,582
Residual Deviance	17,357	17,351	17,339
AIC	17,377	17,371	17,359
Fisher Scoring Iter.	6	6	6

Notes: Standard errors in parentheses. All models use a binomial logistic regression (logit link). Column (1) defines the high-pressure period as the final two minutes of the game. Column (2) adds a rookie kicker indicator and time-based interactions. Column (3) includes playoff effects and their interactions with time and pressure. The three- and four-way interactions are scaled per 100 seconds.

Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$. Abbreviations: C = Close Game, 2_Min = Final Two Minutes, T = Seconds Remaining.

Table 8: Regression Results for Field Goal Success (Logit Model)

Variable	Dependent Variable: Field Goal Success			
	(1) Base Model	(2) Kicker FE	(3) Stadium FE	(4) Full Model
Main Effects				
Intercept	6.49*** (0.21)	6.50*** (0.25)	6.62*** (0.25)	6.64*** (0.29)
Close Game (CG)	-0.19 (0.19)	-0.14 (0.19)	-0.17 (0.19)	-0.13 (0.20)
4th Quarter (Q4)	0.13 (0.21)	0.12 (0.22)	0.11 (0.22)	0.11 (0.22)
Seconds Remaining (SR)	-0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)	0.00 (0.00)
Kick Distance	-0.12*** (0.00)	-0.12*** (0.00)	-0.12*** (0.00)	-0.13*** (0.00)
Kick Distance Squared	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)
Interaction Terms				
CG $\times$ Q4	-0.10 (0.24)	-0.11 (0.25)	-0.10 (0.24)	-0.12 (0.25)
CG $\times$ SR	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Q4 $\times$ SR	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)
CG $\times$ Q4 $\times$ SR (per 100s)	0.06* (0.03)	0.03* (0.01)	0.06* (0.03)	0.06* (0.03)
Model Fit				
Observations	23,582	23,582	23,582	23,582
Residual Deviance	17,357	16,859	17,150	16,766
AIC	17,377	17,235	17,286	17,168
Fisher Scoring Iter.	6	13	6	13

Notes: *Standard errors in parentheses. All models use a binomial logistic regression (logit link). Column (1) is the base model with no fixed effects. Column (2) includes kicker fixed effects. Column (3) adds roof and surface fixed effects. **The coefficient for the three-way interaction term (CG $\times$ Q4 $\times$ SR) has been scaled by 100 to reflect change per 100 seconds. This transformation follows: $\hat{\beta}_{100} = \hat{\beta}_1 \times 100$, and similarly for its standard error.

Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Abbreviations: CG = Close Game, Q4 = 4th Quarter, SR = Seconds Remaining.

Table 9: Effect of Time Pressure in Close Game Situations (Three-Way Interaction Only)

Variable	Dependent Variable: Field Goal Success			
	(1) Base Model	(2) Kicker FE	(3) Stadium FE	(4) Full Model
Main Effects				
Intercept	6.40*** (0.13)	6.46*** (0.18)	6.52*** (0.18)	6.59*** (0.24)
Kick Distance	-0.12*** (0.00)	-0.12*** (0.00)	-0.12*** (0.00)	-0.13*** (0.00)
Kick Distance Squared	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)
Three-Way Interaction (per 100s)				
CG $\times$ Q4 $\times$ SR	0.03** (0.01)	0.03* (0.01)	0.03* (0.01)	0.03* (0.01)
Model Fit				
Observations	23,582	23,582	23,582	23,582
Residual Deviance	17,363	16,864	17,156	16,731
AIC	17,371	17,228	17,280	17,211
Fisher Scoring Iter.	6	13	6	13

Notes: Standard errors in parentheses. All models use a binomial logistic regression (logit link). The three-way interaction term (CG $\times$ Q4 $\times$ SR) is scaled per 100 seconds for interpretability:

$$\text{Rescaled Coefficient} = \text{Original Estimate} \times 100, \quad \text{Rescaled SE} = \text{Original SE} \times 100$$

Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.